

## INFLATION NOW AND THEN

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## INFLATION

*"By a continuing process of inflation, governments can confiscate, secretly and unobserved, an important part of the wealth of their citizens...There is no subtler, no surer means of overturning the existing basis of society than to debauch the currency. The process engages all the hidden forces of economic law on the side of destruction, and does it in a manner which not one man in a million is able to diagnose."*

John Maynard Keynes  
*The Economic Consequence of the Peace (1919)*

The subtle process of inflation, as investors have learned, requires a degree of vigilance to protect against the consequent erosion of purchasing power. History and basic economics have taught us that real returns, not nominal returns, are what matter—real returns being the returns adjusted for inflation. If one cannot generate a return greater than inflation, one's purchasing power erodes inexorably. Thus, being mindful of the effects of inflation is an important aspect of being a successful investor.

### The Complex Interconnections of Inflation

At present, there is a heightened sensitivity to the potential for inflation, given the extraordinary monetary and fiscal policies undertaken by the U.S. and foreign governments to combat the economic crisis. Loose monetary policy, many believe, will result in rampant inflation—the “debauching of the currency” in Keynes's words. However, this is not readily observable from the TIPS (Treasury Inflation Protected Securities) yield curve. In principle, the difference between the yield of traditional Treasury securities and TIPS equals the implied inflation rate. At present, expected inflation does not exceed a rate of 2.6%.

| Implied Inflation Rates |                             |                      |                             |
|-------------------------|-----------------------------|----------------------|-----------------------------|
|                         | US Treasury<br><u>Yield</u> | TIPS<br><u>Yield</u> | Implied<br><u>Inflation</u> |
| 5-Year                  | 2.2%                        | 0.4%                 | 1.9%                        |
| 10-Year                 | 3.5%                        | 1.3%                 | 2.3%                        |
| 30-Year                 | 4.4%                        | 1.8%                 | 2.6%                        |

(as of 5/12/10)

Source: Bloomberg

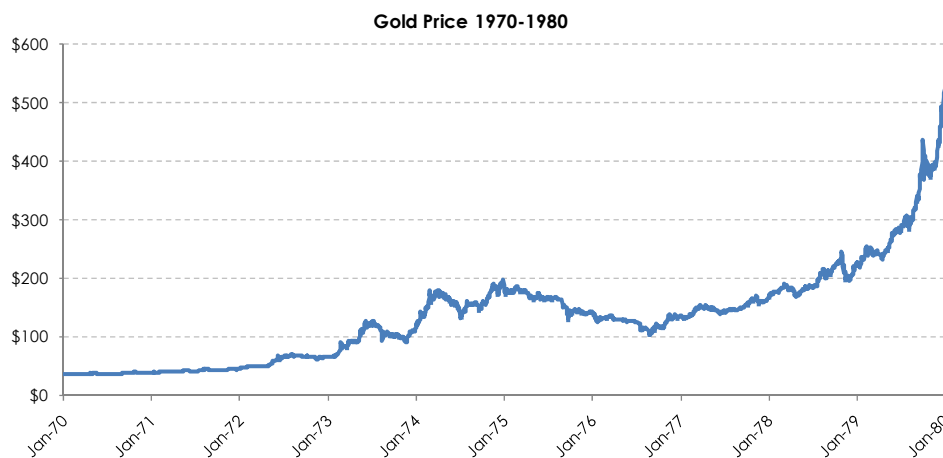
However, investors tend to grossly oversimplify the subject of inflation. For example, the inflation rate normally reported by the financial press is derived from an index of prices—the index being an agglomeration of price increases of a multitude of goods and services. As such, not all goods and services exhibit the same rate of change as the index. Indeed, some prices may actually fall, but it is not necessarily evident by the reported figure. It follows that different constituents of an economy are affected differently by inflation. Consumers may suffer from the phenomenon, whereas natural resource producers may actually benefit. In other words, the effects of inflation are not distributed uniformly throughout the economy and, as a consequence, there are hidden costs to some and benefits to others. Thus, the divergent factions make it difficult to arrive at a consensus for controlling inflation.

In addition, it is unclear which measure of inflation is the proper one to focus upon. Is it the Consumer Price Index ("CPI"), which is a measure of the average price of consumer goods and services purchased by households? Or, is it the Producer Price Index ("PPI")? The PPI measures the average change over time in the selling prices received by domestic producers for their output. Or perhaps it is the Personal Consumption Expenditures ("PCE") price index, a measurement of the average increase in prices for all domestic personal consumption using a variety of data including the U.S. CPI and PPI prices and which is the Federal Reserve's primary measure of inflation. It seems clear that, regardless of the measure, even measuring inflation is a complex task; protecting oneself from its potentially negative effects is an equally difficult, but necessary component of the investment process.

**Predicting the Past**

The complexity of inflation can be illustrated in the experience of two companies, Newmont Mining and Archer Daniels Midland.

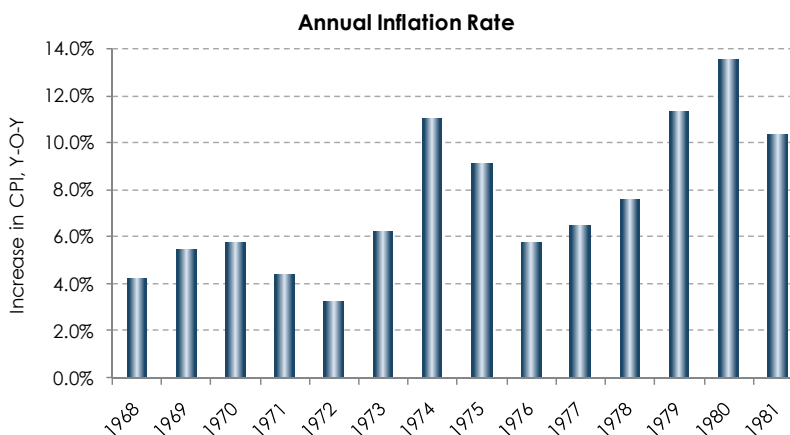
Imagine that we had the opportunity to be transported back to the year 1970. At that time, the price of gold was \$35 per ounce and, through the advantage of hindsight, we know that the decade was one characterized by intense inflationary pressures. Gold, as an inflation hedge, appreciated to well over \$500 an ounce by the beginning of the 1980s.



Cognizant of this, we would pose the following question: Should you have bought Newmont Mining, a gold mining company, in 1970, knowing with certainty the impending inflation?

The short answer, one may be surprised to hear, is “no”; although the Newmont Mining shares appreciated in that period, the investment would not have been as remunerative as one might believe, despite the rate of appreciation in the price of gold. Why is this so? In 1970, Newmont Mining traded at a price-to-earnings ratio of 35x. By inverting the P/E ratio, one would derive the earnings yield, which is an indication of the amount of earnings one receives for a given price paid for the security and which could be compared to prevailing interest rates. In Newmont's case, the earnings yield was 2.8%.

As interest rates rose in response to the inflationary pressures, the earnings yield of Newmont Mining needed to rise as well. However, as the stock's earnings yield rises, basic mathematics dictates that the P/E multiple must fall. Therefore, even though the price of gold increased dramatically, the multiple investors were willing to pay for Newmont's earnings stream declined throughout the decade, negating the contribution of rising gold prices to earnings. Additionally, since Newmont Mining was depleting its gold reserves, the company had to acquire new properties to exploit. However, the replacement cost of the assets also rose with inflation, thus limiting its potential for margin expansion.

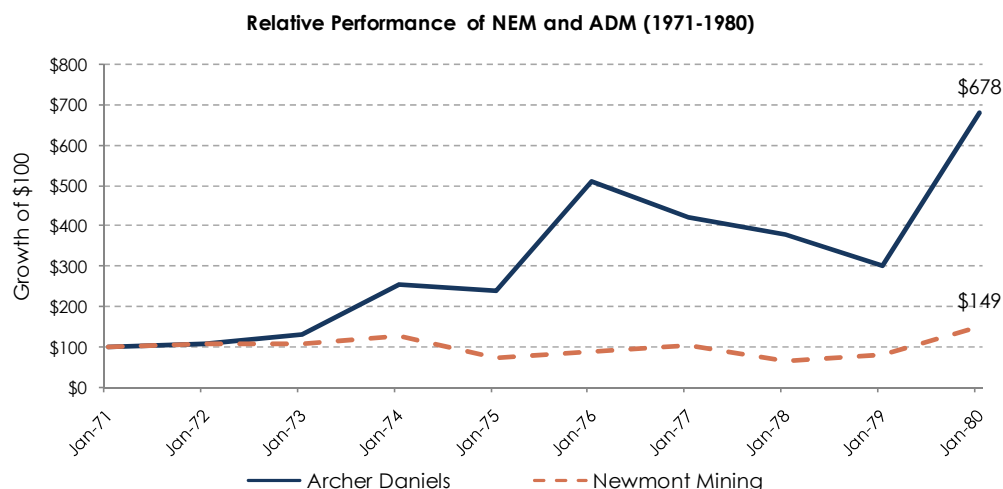


Source: BLS

In contradistinction, Archer Daniels Midland performed much better than Newmont Mining, even though Archer Daniels Midland was not considered to be an obvious inflation beneficiary. What is the business of Archer Daniels? It is a processor of commodities, specifically food products. In this capacity, Archer Daniels mills wheat into flour, corn into cornmeal, etc. The company receives a very narrow margin to produce these products, on the order of 2% of the value of the commodities processed. Therefore, on a bushel of wheat valued at one dollar, the company makes two cents. At two dollars per bushel, though, they make four cents. At higher prices, they can make more money even while the margin remains fixed at 2%.

In 1970, the Archer Daniels price-to-earnings ratio was not far from 10x, or an earnings yield of 10%. In 1980, its price-earnings ratio was also not far from 10x. Consequently, its share price did not suffer from the contraction of valuation multiples that afflicted Newmont Mining, and, therefore, the value of its inflation-generated earnings increases was not negated to any degree by multiple contraction.

Ergo, Archer Daniels Midland was a much more successful investment than Newmont Mining, though perhaps not obvious from the conventional view of the operation of inflation and inflation hedges: the heuristic, "gold company=inflation hedge" did not provide the returns that Newmont Mining shareholders may have expected during that period.



The lesson of Newmont Mining and Archer Daniels Midland should not be lost on the investor—inflation is insidious in that no one really knows what the inflation rate is going to be; even if one did, no one would know what the returns of individual companies would be. If we could not figure out, even retrospectively, that Archer Daniels Midland was a better investment than Newmont Mining, how can we successfully address the effects of inflation on our portfolio prospectively?

### Horizon's Approach

To avoid any confusion, let us state from the outset that Horizon Asset Management does not attempt to predict future inflation rates. To allow predictions of future inflation to dictate investment decisions is fraught with danger, in our view. Firstly, how can anyone make such predictions with any degree of confidence? If it proves to be wrong, such direct “inflation” hedges might limit returns or even cause losses. Secondly, as previously noted, one might get the prediction right, yet still get the investment vehicle wrong.

Rather, our approach towards managing the risk of inflation is to make investments that do not necessarily rely upon the occurrence of inflation to be remunerative, but which would be beneficiaries were it to occur. In this way, the expected outcome is not a binary event. Focusing on a company's underlying ability to produce an adequate return on its invested capital irrespective of the inflationary environment, but which has a positive relationship with inflation, is our preferred method.

### Royalty Companies

A prime example that incorporates this dynamic would be our investments in precious metals royalty companies, such as Franco Nevada and Royal Gold. These types of companies are, essentially, finance companies that purchase interests in gold producers; they do not conduct mining operations themselves. In simple terms, a company such as Franco Nevada purchases a revenue stream derived from the sale of gold by the underlying mining company (e.g., Newmont Mining). The price Franco Nevada pays is equivalent to a discounted present value of the gold that is expected

to be produced. That is, under such a contract, Franco Nevada might pay the current price of \$1,200 per ounce for the first year's production. However, for the second year, it might pay 10% less, or \$1,080 per ounce and \$972 for the third year's output. For the 30<sup>th</sup> year's output, assuming gold prices were to remain constant, Franco Nevada will have paid, in advance, only \$69 per ounce.

This present value amount is typically based on the prevailing gold price. Therefore, if the price of gold were to remain static, and the gold mining company produces the anticipated amount of gold, the rate of return investors in Franco Nevada could expect would be equal to the discount rate applied to the original investment, which in this example is 10%. Note that if one accepts the premise that the price of gold is a proxy for inflation, a scenario of level gold prices would imply the absence of increasing inflation.

Contrast this with purchasing gold outright, either through an ETF or the physical commodity itself. Whereas a flat gold price would produce a zero return in a direct purchase of the commodity, the royalty company would still generate a return equivalent to the discount rate.

If the price of gold were to rise, a holder of physical gold would receive a return equivalent to the rate of appreciation. However, the royalty company would also receive the appreciated gold price in addition to the discount rate, assuming no commodity hedges are in place.

What would happen if the price of gold falls? An investor in physical gold would experience a loss in direct proportion to the price decline. For example, if gold were to fall from the current level of approximately \$1,200 per ounce to \$1,080 per ounce, a loss of 10% would be incurred. However, for the royalty company investor, the decline in gold prices would be offset by the discount rate of the future cash flows. Thus, in principle, if a royalty company could purchase production output at a discount of, say, 10% over 30 years it could actually tolerate a 10% decline in gold prices each and every year over a 30-year period without sustaining a loss. This is simply not possible through a direct investment in the precious metal.

### *Spread-Based Businesses*

Another type of investment which could prosper in an inflationary environment would be companies that are able to pass through the costs of inflation through a spread-based business model. These are defined as companies that charge a fixed percentage fee for their products or services. This would include companies such as the aforementioned Archer Daniels Midland or MasterCard. MasterCard is interesting in that it is usually erroneously categorized as a financial institution, similar to banks and other companies that lend money. In reality, MasterCard is simply a processor: it does not issue credit cards, it simply processes the transactions made on credit cards bearing its brand.

The company's primary source of revenue comes from charging a small percentage of a transaction's dollar value as a fee for providing the processing service. Therefore, for illustrative purposes, let us say that MasterCard's fee amounts to 1% of the dollar value of transactions, so that a purchase of \$100 would generate a fee of \$1 (in reality, MasterCard's fee averages about 10 basis points, or 0.10%, but is applied to trillions of dollars in transactions). If a product's price were to increase by, say 10%, care of inflation, the total dollar value of the purchase would be \$110, with a proportionate increase in the fee to \$1.10. Thus, a company such as MasterCard, which is an operator of a spread-based business, is immunized from inflation. Indeed, it might actually benefit from a general rise in the price of goods and services; in this simplified example, the 10% increase in revenue requires no direct increase in expense, such as from the number of personnel, computer processing capacity or office space and, so, should be unusually profitable.

### *Real Estate*

Although the real estate industry has experienced much pain during the recent economic crisis, it has not invalidated real estate as an inflation beneficiary. It has been Horizon's practice to concentrate on select commercial real estate companies having strong balance sheets and unique income-producing properties, such as Vornado Realty and Forest City Enterprises. Longtime clients may recognize these companies as being core holdings at the firm for many years. Aside from their fundamental investment merits, these companies provide an inherent hedge against the effects of inflation through multiple modalities. For example, inflation increases the value of the underlying real estate properties, so that the net asset value rises in inflationary periods. Additionally, as their commercial leases come up for renewal, higher prices can be charged to reflect the effects of inflation on market rates. Finally, real estate companies typically utilize leverage to purchase their properties. To the extent that the mortgage payments are fixed, their debt will be paid back with cheaper dollars. In this sense, leveraged companies in general can be beneficial to an investment portfolio in inflationary periods.



**Conclusion**

Unlike a stock market crash, inflation erodes one's wealth silently and covertly. Given the massive amounts of stimulus provided by the U.S. government over the past several years, one is well-advised to remain vigilant in protecting one's assets from the effects of inflation. However, the traditional ways to do so, such as purchasing physical gold, may not be the most effective. We believe that better alternatives exist—business models that can protect a portfolio during periods of inflation, but which do not necessarily depend upon the manifestation of inflation in order to be profitable. Although this essay is not intended to be an all-encompassing discussion on the subject, we hope that it has provided you with a better understanding of the issues pertaining to inflation and our approach towards managing the related risks. Should you wish to discuss this further, please do not hesitate to contact us.

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